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Sensitivity Analysis of Particle Tracking Algorithm for Event Based Velocimetry

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Building on recent validation studies, the present work performs a systematic sensitivity analysis of a weighted event-based tracking algorithm applied to two configurations. The first is freestream and post stall flow around an SD7003 wing in a water tunnel. The second is a freestream-cavity arrangement that combines fast outer flow with slow, highly sheared recirculation. The study quantifies how event camera bias settings, time segment duration, and internal tracking thresholds influence event density, computational cost, and velocity accuracy. Results show that high pass filtering and data volume primarily control computation time, whereas time segmentation and the minimum ellipse diameter have a stronger impact on the recovered mean velocity. In the slow cavity, temporal thresholds that govern inter event gaps and time linearity become the dominant factors. The analysis yields practical guidelines for choosing camera biases and tracking thresholds so that reliable velocity estimates can be obtained in both fast and slow regions of a flow while maintaining manageable data rates and processing times.

Nomenclature

$I(t)$	= Intensity of each pixel	EB − PIV	= Event-Based PIV
$C_{Temporal}$	= Temporal Contrast	HPF	= High Pass Filter
α	= Angle of Attack	A_{Thr}	= Accumulation Threshold
x, y	= Co-ordinates in X and Y directions	$D_{Ellipse}$	= Minimum Ellipse Diameter
U, V	= Velocities in X and Y Directions	T_{Thr}	= Time Threshold
c	= Wing Chord	B_{Limit}	= Base Distance Limit
U_∞	= Freestream Velocity	TL_{Thr}	= Time Linearity Threshold
PIV	= Particle Image Velocimetry	θ_W	= Angular Weight

I. Introduction

Particle Image Velocimetry (PIV) remains a standard technique for measuring planar flow fields in experimental fluid dynamics and is widely used to characterize complex and turbulent flows [1]. Temporally resolved PIV requires high speed cameras and lasers and requires substantial data storage and transfer requirements without a proportional gain in velocity information content [2]. These constraints limit the accessible temporal dynamic range, or the duration of data acquisition, especially when high sampling rates are needed in unsteady or high-speed flows.

Event-based cameras offer an alternative sensing paradigm that can mitigate some of these limitations.

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Instead of recording full image frames at a fixed rate, they asynchronously report pixel wise changes in logarithmic light intensity once a user defined contrast threshold is exceeded [3,4]. This architecture provides microsecond scale timestamp precision and produces sparse data streams in scenes where only tracer particles move, which is particularly attractive for velocimetry since the static dark background is largely absent from the data [3].

Applications of event cameras to velocimetry have followed two main directions. The first is event-based particle tracking velocimetry in which individual tracer trajectories are reconstructed directly from the event stream. Drazen et al. [5] demonstrated proof of concept real time particle tracking by comparing an event camera with a high-speed CMOS camera in a simple freestream configuration. Subsequent studies extended this concept to three-dimensional flow reconstruction with multiple cameras, showing that volumetric particle trajectories and velocity fields can be recovered in more complex geometries [6–8]. The second direction adapts cross correlation PIV methods by reconstructing pseudo images through temporal accumulation of events. Willert and Klinner [9] introduced event-based imaging velocimetry and evaluated continuous wave illumination for planar flow measurements using contrast based accumulation within interrogation windows [9]. Willert [10] later used pulsed laser illumination to characterize event latency and bandwidth limits and proposed acquisition strategies that avoid readout saturation while preserving velocity accuracy.

In parallel with these hardware and acquisition developments, several event-based processing algorithms have been proposed for planar velocimetry. Gunasekaran et al. [11] compared different tracking and fitting strategies for event streams and showed that nonlinear regression along particle tracks can provide velocity estimates that agree with reference PIV while using significantly fewer data. AlSattam et al. [12] developed KF PEV, a causal Kalman filter based particle event velocimetry method that updates particle position and velocity estimates sequentially as new events arrive and demonstrated improved robustness in noisy and unsteady flows. The filter design follows standard state space modeling and measurement update principles but is adapted to the asynchronous timing of event cameras and the sparse sampling of individual particle tracks [13].

More recently, several studies have focused on quantifying the practical performance limits of event-based velocimetry under controlled conditions. Cao et al. [14] used a digital micro mirror device to generate synthetic particle fields and benchmarked event-based imaging velocimetry accuracy over a range of seeding densities, accumulation times, and event rates. Franceschelli et al. [15] applied EBIV to turbulent flows and showed that event-based measurements can provide efficient low dimensional representations of dominant flow structures that are suitable for reduced order modeling. Maya et al. [16] constructed performance maps for event cameras used in PIV and showed that bandwidth and latency impose coupled constraints on allowable event rate, accumulation time, and particle image density, which in turn bound the achievable velocity dynamic range and uncertainty of EBIV measurements.

Despite this growing body of work, the accuracy and robustness of event-based velocimetry remain strongly dependent on both sensor configuration and algorithm parameters. Camera bias settings such as high pass filtering, contrast thresholds, and refractory period influence event density, polarity ratio, and timing, while tracking and filtering algorithms introduce additional thresholds that govern track formation, validation, and velocity estimation. In particular, Kalman filter based approaches such as KF PEV require careful tuning of process and measurement noise covariances and rejection criteria to maintain causal operation without sacrificing accuracy [12,13]. The present study extends previous validation efforts by performing a systematic sensitivity analysis of the weighted tracking algorithm applied to different flow fields. The accuracy of velocity estimate, event data volume, and computational cost depend on camera bias settings, time segment duration, and key thresholds in the tracking algorithm. The resulting guidelines are intended to assist practitioners in configuring event-based velocimetry systems for specific aerodynamic applications and in balancing measurement fidelity against data and processing constraints.

II. Weighted Tracking Algorithm

This study focuses on the weighted tracking method described in detail in [17], which forms the basis for both linear and nonlinear trajectory fitting. Events are first filtered to retain only positive polarity and to suppress noise using inceptive event and polarity filters. The filtered events are then grouped into tracks based on spatial proximity and temporal continuity. Each track is formed by linking successive events that fall within a specified spatial window and

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time interval. After the tracks are assembled, particle velocities are estimated from the spatial (x, y) and temporal (t) coordinates using either a linear or a nonlinear fit. Nonlinear interpolation improves accuracy in regions with particle acceleration or curved trajectories. A weighting scheme based on event accumulation, or track strength, is applied to reduce sensitivity to noise and to suppress low confidence tracks. The performance of this procedure depends on several settings in the camera, in the post processing, and in the internal thresholds of the algorithm, all of which can influence the final velocity estimates. The specific parameters examined in the present sensitivity analysis are introduced in the next section.

III. Sensitivity Parameters

The performance of event-based particle tracking algorithms depends not only on the tracking methodology but also on sensor-level configurations and algorithmic threshold parameters. Two key categories of settings influence the sensitivity and accuracy of the system. The first category consists of camera-level bias settings that control the behavior of the event sensor. The second category consists of algorithm-level thresholds that govern event segmentation, track formation, track validation, and velocity estimation.

3.1 Event Camera Bias Settings

Prophesee event cameras such as the EVK4 offer configurable bias parameters that modulate how individual pixels detect and respond to changes in light intensity. These biases control the sensitivity, frequency content, and signal quality of the events triggered by intensity changes. For particle velocimetry applications, five bias settings are most relevant.

- **bias_diff_on** sets the contrast threshold for ON events. This parameter determines how much brighter a pixel must become to trigger a positive event. Increasing this value reduces sensitivity to positive brightness changes, which leads to fewer ON events and a lower noise level.
- **bias_diff_off** sets the contrast threshold for OFF events. This parameter determines how much darker a pixel must become to trigger a negative event. As with bias_diff_on, increasing this value reduces sensitivity to negative changes and can suppress background noise.
- **bias_hpf** adjusts the high pass filter at the pixel level. A higher value filters out slowly varying signals so that only rapid changes, such as those produced by moving particles, generate events. This setting is particularly effective in removing static background or ambient lighting variations.
- **bias_fo** controls the low pass filter response and limits the ability to detect very rapid changes. Lowering this value makes the sensor more sensitive to high frequency variations but can increase flicker induced noise.
- **bias_refr** sets the refractory period, which is the minimum time a pixel must wait before generating another event. Increasing this value reduces the likelihood of duplicate events from a single intensity change and limits the overall event rate.

These parameters interact in complex ways and influence the density of events, the ratio of ON to OFF polarity, and the effective timing resolution of the event stream. For example, increasing the high pass filter setting can suppress background activity but may also reduce sensitivity to slow moving particles. In this study, the sensitivity analysis focuses primarily on variations in the high pass filter, the ON contrast threshold, and the refractory period in order to quantify their impact on tracking stability and velocity field reconstruction.

3.2 Threshold Settings in the Weighted Tracking Algorithm

The weighted tracking algorithm includes several internal thresholds and parameters that directly influence the construction of particle tracks and the resulting velocity estimates. These settings govern both the temporal segmentation of events and the logic for grouping them into coherent particle trajectories. A visual representation of some of the thresholds is shown in Figure 1.

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1. Inceptive Event (IE) filter window: This filter discards noise by identifying polarity paired events within a specified time window. A typical range for the inceptive window is 3000 to 10000 µs. Events that do not have an opposite polarity partner within this window are rejected.

2. Time segmentation window: The event stream is segmented into fixed duration time bins, for example 10000 µs, before applying the tracking algorithm. Shorter windows increase temporal resolution but can reduce particle visibility and track continuity. Longer windows improve track continuity in sparse regions but can blend distinct flow features that occur within the same segment.

3. Track formation thresholds: The core tracking step uses elliptical gating to assign events to existing tracks. Several parameters control this association rule.

- **Time Threshold, T_{Thr}** is the maximum allowed time gap between successive events in a track, for example 3000 µs. It limits how long a track can persist without receiving new events.
- **Minimum Ellipse Diameter, $D_{Ellipse}$** is the minimum ellipse dimension required to initiate tracking, typically on the order of 1.8 pixels. It sets a lower bound on the search region size.
- **Steps** is the number of discrete scaling levels used as the track grows.
- **Scale** controls how quickly the elliptical search region expands with track length, which in turn affects how aggressively nearby events are merged into the same track.
- **Angular Weight, θ_W** is an angular weighting factor used during direction estimation. As a track ages the weighting shifts from recent events toward the historical direction to stabilize long tracks.

Together these parameters control the spatial tolerance and directional sensitivity of the track formation process and are especially important in regions with strong velocity gradients or high noise.

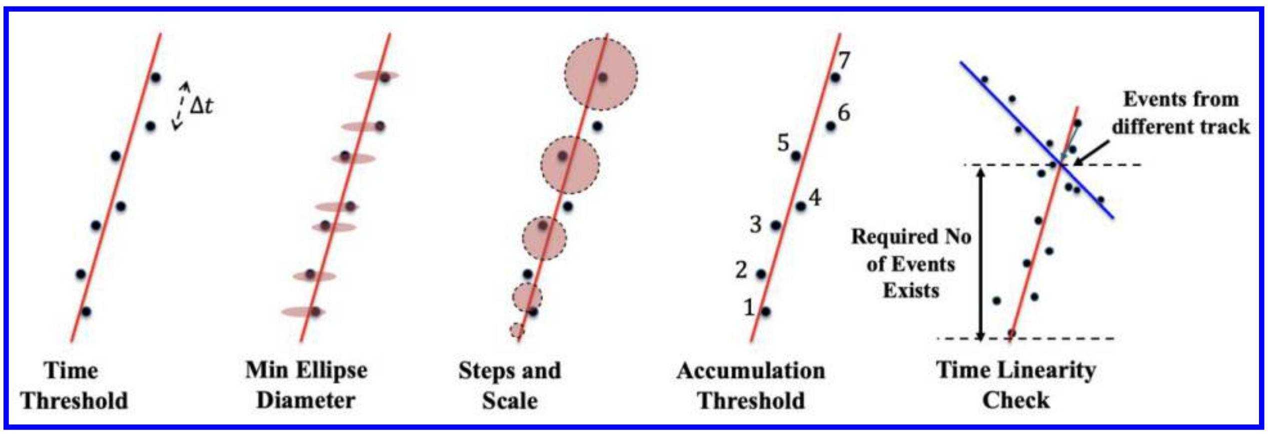

Figure 1: Visual Depiction of Threshold Setting

4. Track Validation Thresholds: Once a track has been formed, the algorithm then performs a validation check on all the tracks to determine which ones are reliable for velocity estimation or "valid". The following thresholds are critical:

- **Accumulation Threshold (A_{Thr}):** After an event stream has been processed and provisional tracks have been formed, the total number of events in each track is computed and propagated back to all events in that track. Tracks with fewer than a specified number of events, typically on the order of 5 to 10, are discarded. This removes extremely short tracks and helps suppress weak or spurious tracks, especially in low seeding conditions.
- **Time Linearity Check (TL_{Thr}):** For tracks that survive the accumulation filter, a time linearity test is applied to the event timestamps. A variance-based change detection function is used to identify segments of nearly uniform inter event timing. The length of the longest time linear segment is compared against a threshold such as 10 events. Tracks with a longest segment shorter than this threshold are rejected. This step removes

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bursty or irregular tracks that do not follow the expected quasi uniform timing associated with a particle moving through the field of view and is particularly important in flows with strong temporal gradients or in noisy recordings.

IV. Experimental setup

Two experimental data sets were used for the sensitivity studies in this work. The influence of camera bias settings and time thresholds was examined using an event based particle tracking velocimetry data set for a simple freestream and for flow over a wing. The sensitivity of the particle tracking algorithm threshold settings was evaluated using data from a freestream cavity experiment. These two experiments are described below.

1. Experiments Used for Sensitivity Studies on Camera HPF Settings and Time Segmentation

The sensitivity analysis of camera bias settings and time segmentation was carried out using data obtained in Gunasekaran et al. [11], where event-based particle tracking velocimetry was applied to freestream flows and to flow over an SD7003 wing in the University of Dayton water tunnel. This facility includes a closed loop water tunnel with a transparent test section that is 38 cm wide and 46 cm high, with a total length of 152 cm.

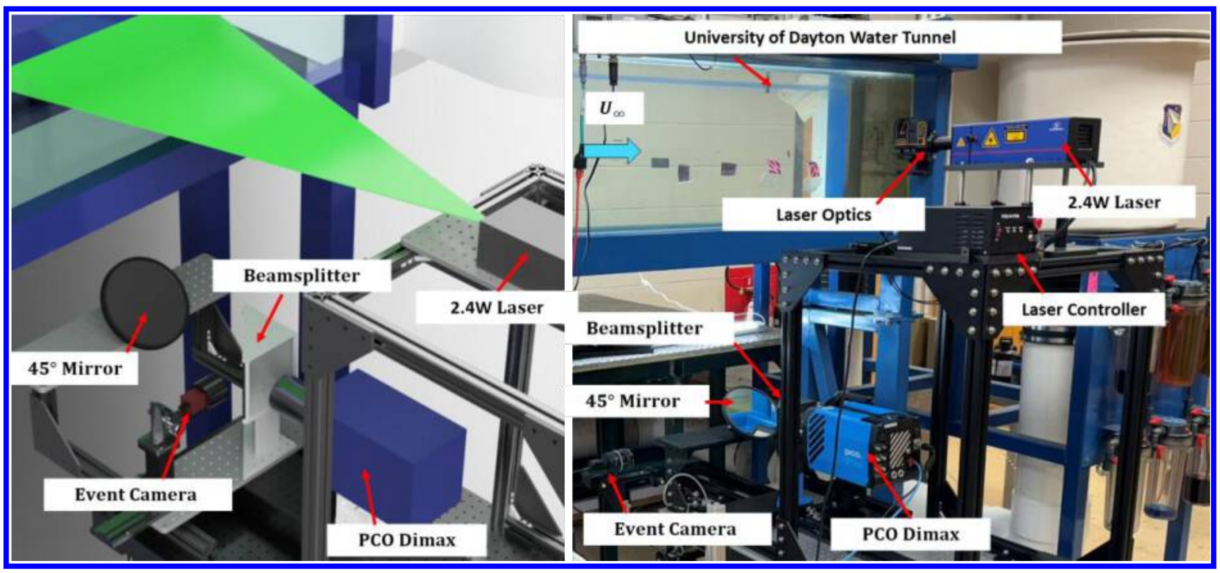

Figure 2 A schematic (left) and actual setup (right) of the event-based and traditional velocimetry system in the University of Dayton Water Tunnel.

The first set of measurements consisted of freestream experiments without the wing installed. In this configuration, the water tunnel was operated at three different freestream velocities of 0.2 m/s, 0.3 m/s, and 0.37 m/s. These conditions were chosen because the bulk velocity could be determined independently from the tunnel operating conditions and from reference PIV, which provided a baseline for evaluating how accurately event-based velocimetry recovers the true flow speed. The freestream data are particularly useful for sensitivity studies, since the underlying flow is nominally uniform and any observed variations in the estimated velocities can be attributed primarily to sensor configuration, processing parameters, or algorithm thresholds rather than to spatial variations in the flow.

For the second set of measurements, a transparent SD7003 wing was mounted in the test section and the flow around the wing was recorded at a freestream velocity of 0.28 m/s at various angles of attack. Illumination was provided by a continuous laser light sheet aligned with the midspan plane of the wing. The flow was seeded with neutrally buoyant polyamide tracer particles. A Prophesee EVK4 event-based camera equipped with a 50 mm lens was mounted below the test section, capturing events at a spatial resolution of 7900 pix/m. Further details on the tunnel, seeding density, optics, and calibration procedures are shown in [11]. A schematic of the setup used in shown in Figure 2.

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2. Experiments Used for Sensitivity Studies on Weighted Tracking Algorithm Threshold Settings

Sensitivity studies were also performed using data obtained from Vincent et al. [21]. In this second application, the weighted particle tracking algorithm was applied to a freestream cavity configuration that was designed as a physical reservoir computer. In this configuration, a cavity embedded in a wing is subjected to upstream vortical disturbances. As shown in Figure 3, the setup consists of a wing with an open cavity while a pitching upstream wing generates a periodic vortical gust that impinges on the cavity. The goal of this effort is to experimentally characterize how the cavity flow encodes the history of the incoming gusts and to assess its suitability as a reservoir for flow-based computing.

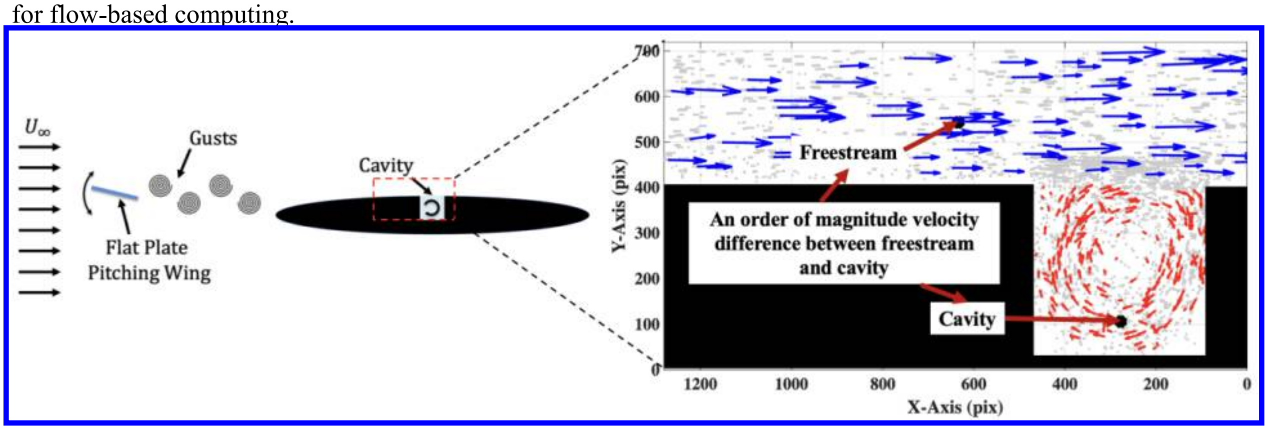

Figure 3 Schematic of freestream–cavity experiment showing representative event-based velocity field with high-speed freestream and low-speed cavity recirculation regions.

From a measurement standpoint, this configuration presents several challenges for conventional PIV. The shear layer that spans the cavity opening exhibits large velocity gradients. The characteristic velocities inside the cavity are one to two orders of magnitude smaller than the freestream velocity. In addition, the flow must be recorded continuously over several minutes in order to reveal long time scale correlations between the gust input and the cavity response. These requirements make it difficult to choose a single Δt for PIV that is simultaneously appropriate for both the high speed freestream region and the much slower cavity flow. Because events are triggered asynchronously by intensity changes, an event based camera captures both the high speed freestream structures and the slower recirculating motions inside the cavity without the need to tune the sampling interval separately for each region.

This data set also provides an opportunity to test the algorithm's performance in two very different flow fields that appear within a single field of view. In the freestream region, the flow is nominally uniform and steady and does not exhibit strong spatial or temporal gradients. In contrast, the flow inside the cavity features large velocity gradients, curved streamlines, pronounced temporal dependence, and characteristic velocities that are one to two orders of magnitude smaller than those in the freestream. By applying similar threshold settings to these two distinct regions, the performance of the weighted tracking algorithm can be evaluated under flow conditions that differ strongly in both speed and complexity while sharing the same threshold settings.

For the purpose of the sensitivity analysis in the present study, the influence of the various algorithm thresholds is examined using a steady operating condition without the gust interaction. The mean velocity magnitude in the freestream and cavity regions is compared across different threshold settings in order to quantify how the velocity estimates in each region change as the internal parameters of the weighted tracking algorithm are varied.

V. Sensitivity Study of Camera HBF Settings and Time Segmentation on Freestream Data

5.1 Global Sensitivity to High-Pass Filter (HPF) Bias

To evaluate how sensor level filtering influences event generation and tracking quality, a sensitivity study was conducted on the camera high pass filter bias. Three HPF bias values of 80, 100, and 120 were applied during separate data acquisitions, while the low pass filter bias, was held at its minimum setting of -35 to maximize responsiveness to high frequency variations. As shown in Figure 4 for a freestream only flow field at 0.3 m/s, increasing the HPF value progressively removes slower illumination changes and produces a sparser event field. At a bias_hpf of 80, shown in Figure 4a, the event density is high and yields a detailed representation of the particle field, but it also includes a significant number of background and low motion events. When the HPF is increased to 100, Figure 4b, much of the background activity is suppressed and the particle related features become more distinct and spatially coherent. At 120 HPF as shown in Figure 4c, the event stream becomes even sparser and retains primarily the fastest moving particle signatures. This setting minimizes noise and enhances contrast for high-speed features, but it also increases the risk of losing valid particle tracks in regions with slower motion. These observations highlight the tradeoff between event density and signal to noise ratio when tuning the HPF bias. Optimal performance requires a balance that removes background noise yet preserves enough particle events to support accurate velocity reconstruction.

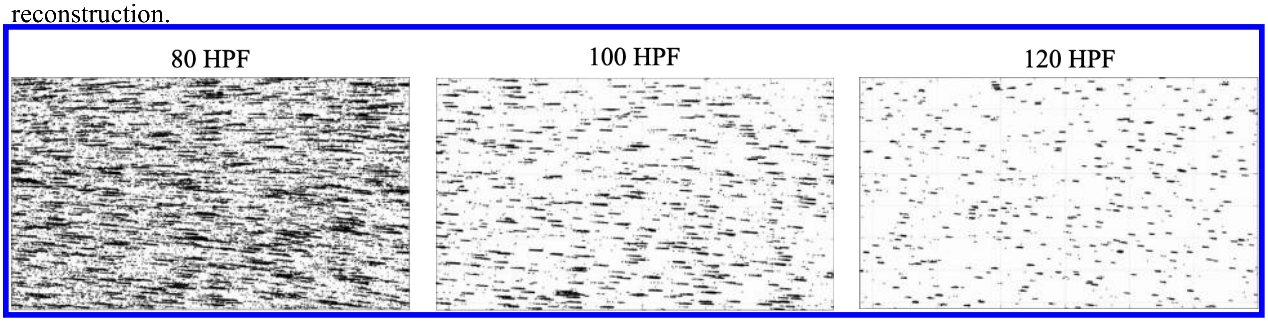

Figure 4 Event density comparison for HPF biases of 80, 100, and 120. Higher HPF reduces background noise but also decreases overall event count.

5.2 Sensitivity to Time Segment Duration

The influence of temporal segmentation on event density and flow reconstruction was investigated by varying the time window length used during processing. For each case, a total of 4 seconds of event data were recorded and then divided into shorter segments with durations of 0.01 s, 0.02 s, 0.03 s, and 0.04 s. As shown in Figure 5, increasing the time segment duration leads to a higher accumulation of events within each segment and produces denser visual patterns. Shorter segments such as 0.01s provide finer temporal resolution, which helps preserve transient flow features and reduces overlap between events. However, the reduced number of events in each segment can result in fragmented tracks and limited spatial coverage, especially in low seeding conditions or at higher flow speeds. In contrast, longer durations such as 0.04s yield much denser event fields, which improves tracking stability and spatial coherence but introduces temporal blur and increases the likelihood of particle path crossings. This trade off emphasizes the need to select a time segment duration that balances spatial coverage, event accumulation, and temporal fidelity, particularly in unsteady or spatially complex flows.

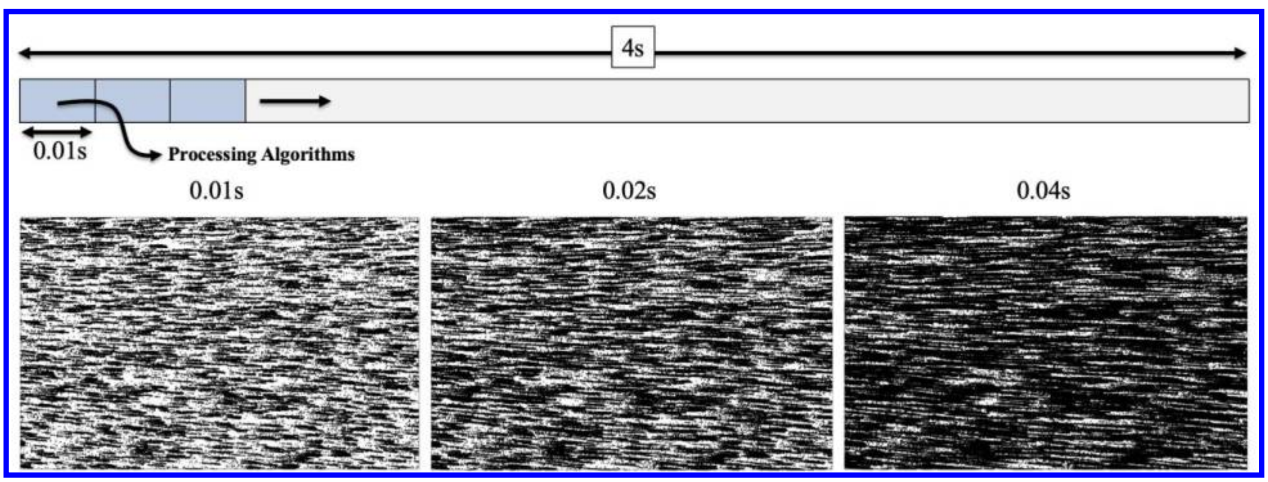

Figure 5 Effect of time segment duration on event density. Segments of 0.01 s, 0.02 s, and 0.04 s show increasing event accumulation and reduced temporal resolution.

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5.3 Computational Time Sensitivity to HPF Bias and Data Volume

To assess the computational cost associated with different event processing configurations, the total processing time was evaluated as a function of time segment duration and percentage of data used under varying high pass filter (HPF) bias settings. Figure 6 presents contour plots of computational time in minutes for HPF values of 80, 100, and 120 across three flow speeds of 0.2 m/s, 0.3 m/s, and 0.37 m/s. The results reveal a clear and consistent trend. Higher HPF bias settings substantially reduce computational time because the sparser event streams produced by larger HPF values suppress low frequency and background activity. At a bias_hpf of 80, the computational burden is significantly higher, particularly when longer time segments and larger fractions of the data set are processed. As the HPF is increased to 100 and 120, the event density decreases, which leads to fewer particle tracks and shorter overall processing durations.

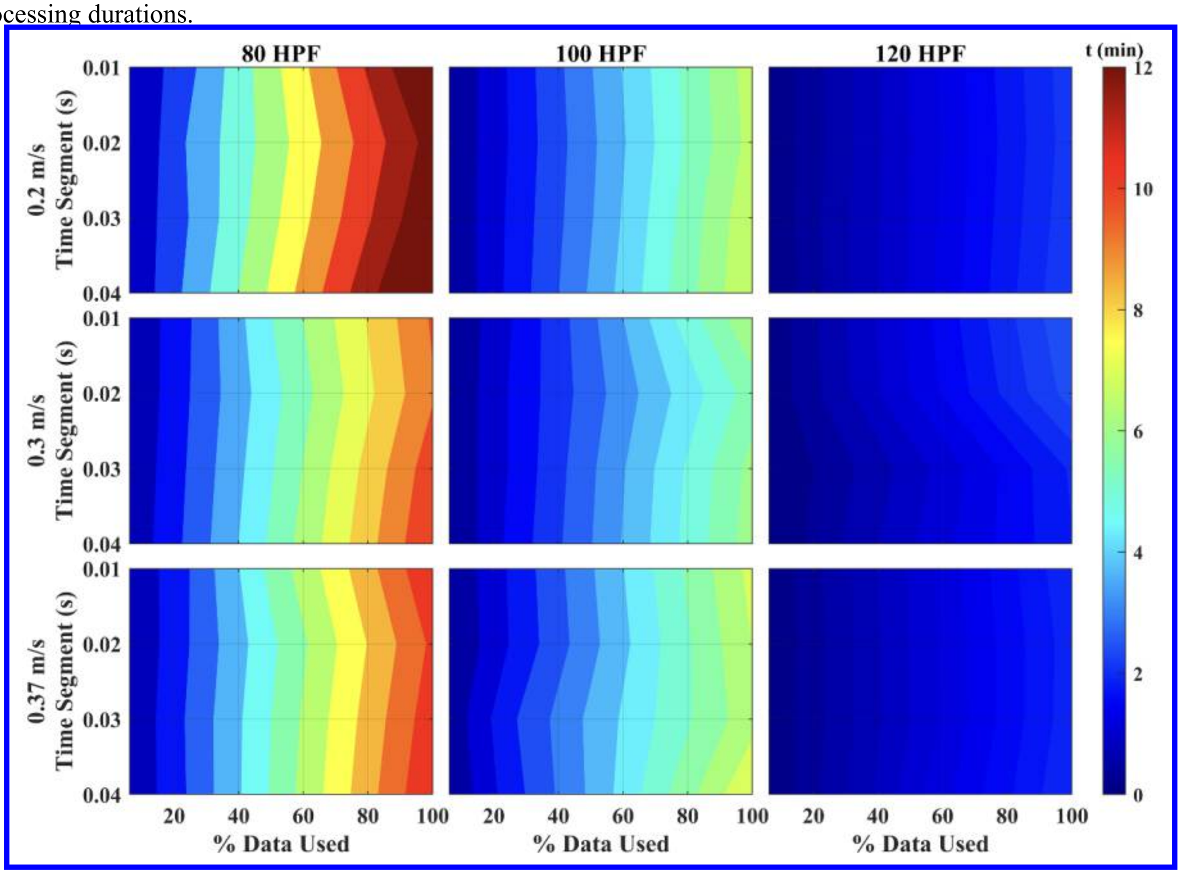

Figure 6 Computational time contours as a function of time segment duration and percentage of data used for HPF biases of 80, 100, and 120. Higher HPF values significantly reduce processing time.

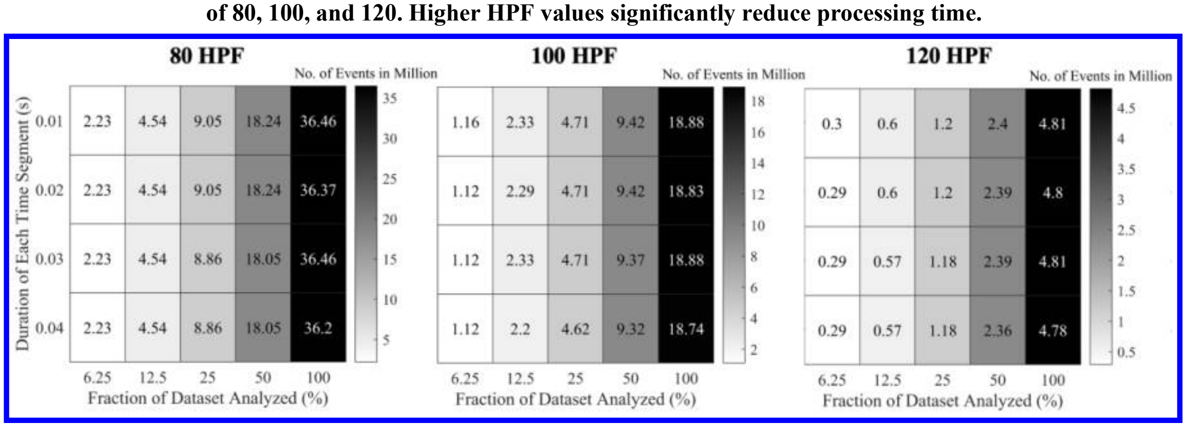

Figure 7 Total number of events (in millions) as a function of time segment duration and dataset fraction for HPF biases of 80, 100, and 120. Higher HPF settings significantly reduce event volume.

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The total number of events processed plays a critical role in determining the computational burden of event-based tracking algorithms. Figure 7 presents the total event count (in millions) as a function of both time segment duration and fraction of dataset analyzed, for three different HPF bias settings. As expected, event volume is inversely related to HPF bias. At bias_hpf = 80, the lowest filter setting, the number of events reaches up to 36 million for full dataset usage with 0.01–0.02 s segments. In contrast, the event volume drops significantly to ~18 million at bias_hpf = 100, and further down to <5 million at bias_hpf = 120. These reductions align closely with the observed trends in computational time (Figure 6), confirming that event density is the dominant factor in processing duration. Interestingly, changes in time segment duration have little impact on total event count for a given fraction of the dataset, suggesting that event generation is temporally uniform under constant flow conditions. These findings reinforce the role of sensor-level filtering (HPF) as a powerful tool not only for noise suppression but also for managing processing load in real-time applications.

4. Sensitivity of Median Velocity to Time Segment Duration and HPF Bias

The impact of HPF bias, time segment duration, and data volume on the estimated median streamwise velocity was examined. Figure 8 presents contours of normalized median velocity for combinations of HPF settings of 80, 100, and 120, time segment durations from 0.01 s to 0.04 s, and data usage between 20 percent and 100 percent.

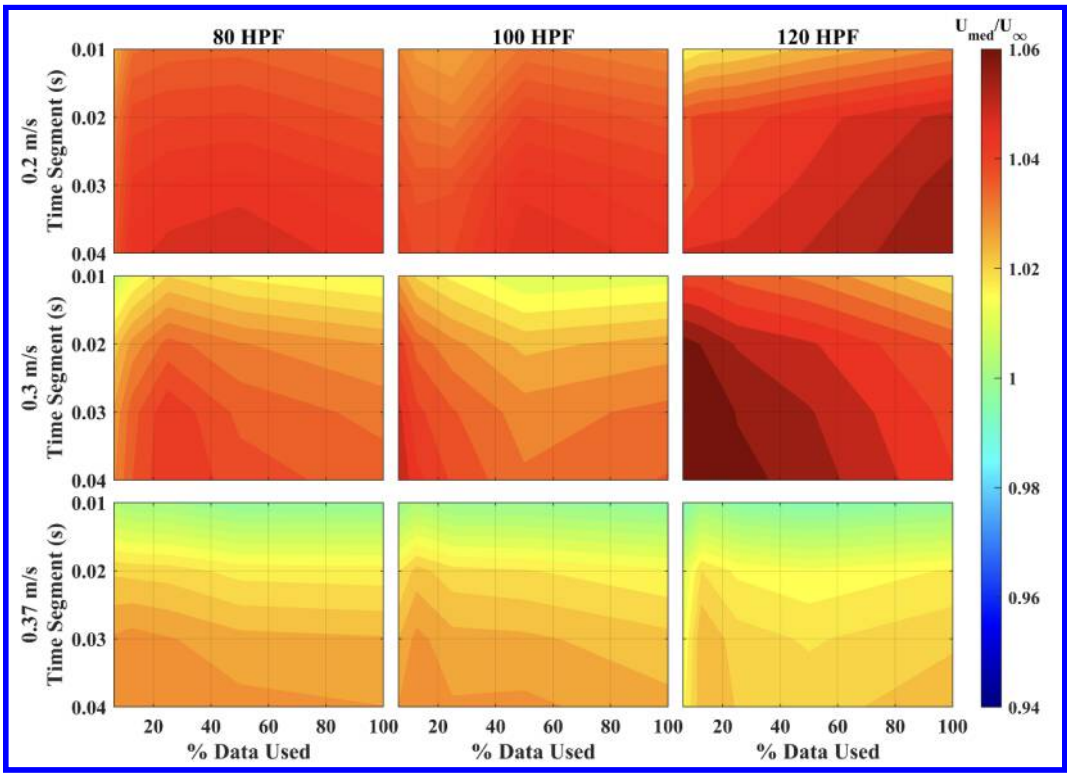

Figure 8 Normalized median U-velocity as a function of time segment duration, data volume, and HPF bias. Velocity estimates are more sensitive to segment duration than data percentage.

The results indicate that time segment duration has a stronger influence on the velocity estimates than the percentage of data used. Across all HPF values and flow speeds, shorter segments of 0.01 s and 0.02 s tend to slightly underestimate the velocity, while longer segments produce larger values, particularly when combined with higher HPF biases. This trend is consistent with the behavior of the nonlinear fitting algorithm applied to predominantly linear particle tracks, since longer segments increase event density along each track and can introduce a small curvature bias in the fitted trajectories. In contrast, the effect of data volume is relatively minor. Once a sufficient fraction of the data is used, the median velocity estimates remain nearly constant as the percentage of data increases. This suggests that, after a critical event density is reached, additional data contributes little to improving velocity accuracy. These results highlight the importance of carefully tuning the time segmentation, especially when nonlinear track fits are employed, in order to avoid systematic bias in the reconstructed flow measurements.

5. Trade-Off Between Event Volume and Velocity Accuracy at 10° AOA

This analysis examines the balance between event data volume, computational efficiency, and velocity estimation accuracy at an angle of attack of 10°. Figure 9 shows the computational time in the top row and the normalized median velocity in the bottom row for three HPF bias settings of 80, 100, and 120, evaluated over different time segment durations and percentages of data analyzed. The plots demonstrate that nearly identical velocity estimates can be obtained from configurations that differ significantly in computational cost. For instance, using the full data set with a low HPF setting of 80 produces the longest processing time, exceeding 30 minutes. In contrast, a much leaner configuration that uses only 20 percent of the data with a higher HPF setting of 120 reduces the processing time to less than 5 minutes, yet the resulting median velocity is effectively unchanged. These results show that increasing the HPF bias and limiting the data volume can substantially reduce computational effort without degrading velocity accuracy, provided that the time segment duration and track quality remain adequate. This trade off provides a basis for selecting practical operating points in event-based flow diagnostics where both efficiency and fidelity are maintained.

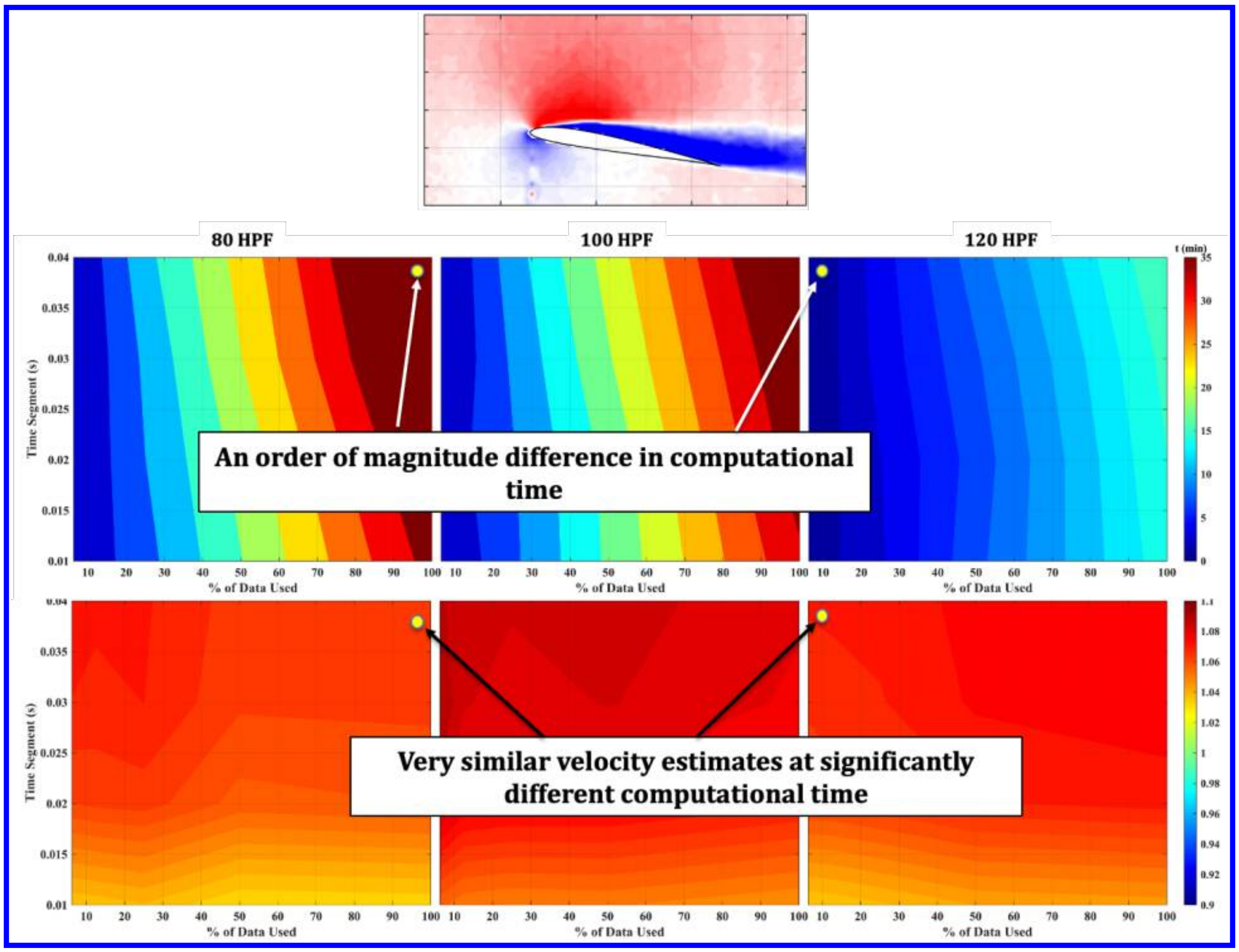

Figure 9 Comparison of computational time (top) and normalized median velocity (bottom) at 10° AOA across HPF settings. Similar velocity estimates are achieved with drastically different processing times.

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VI.Sensitivity Study of Tracking Algorithm Thresholds

In addition to camera level bias settings and temporal segmentation, the performance of event-based velocimetry depends on the thresholds that govern how event streams are assembled into tracks and how those tracks are judged to be physically meaningful. The tracking algorithm is implemented as a two-step process. In the first step, individual events are grouped into candidate trajectories using an adaptive elliptical search. In the second step, each candidate trajectory is subjected to a validation procedure that checks whether it satisfies prescribed temporal and spatial requirements before it is used for velocity estimation. The following sections describe in detail how each threshold contributes to the processing of a potential track and how these choices affect the resulting velocity fields. The freestream-cavity dataset is used for this analysis.

A. Velocity Sensitivity of Track Defining Thresholds

In an ideal situation the weighted tracking algorithm would return the same mean velocity for every combination of thresholds and each panel in Figure 10 would appear as a single uniform color. The top row shows that this is not the case in the freestream region. In the left panel the mean velocity varies strongly with $D_{Ellipse}$ while it is only weakly affected by accumulation threshold, A_{Thr}. This behavior follows directly from the way $D_{Ellipse}$ enters the implementation. $D_{Ellipse}$ sets the initial size of the elliptical search region that controls how much the ellipse can elongate along the direction of motion as the track grows.

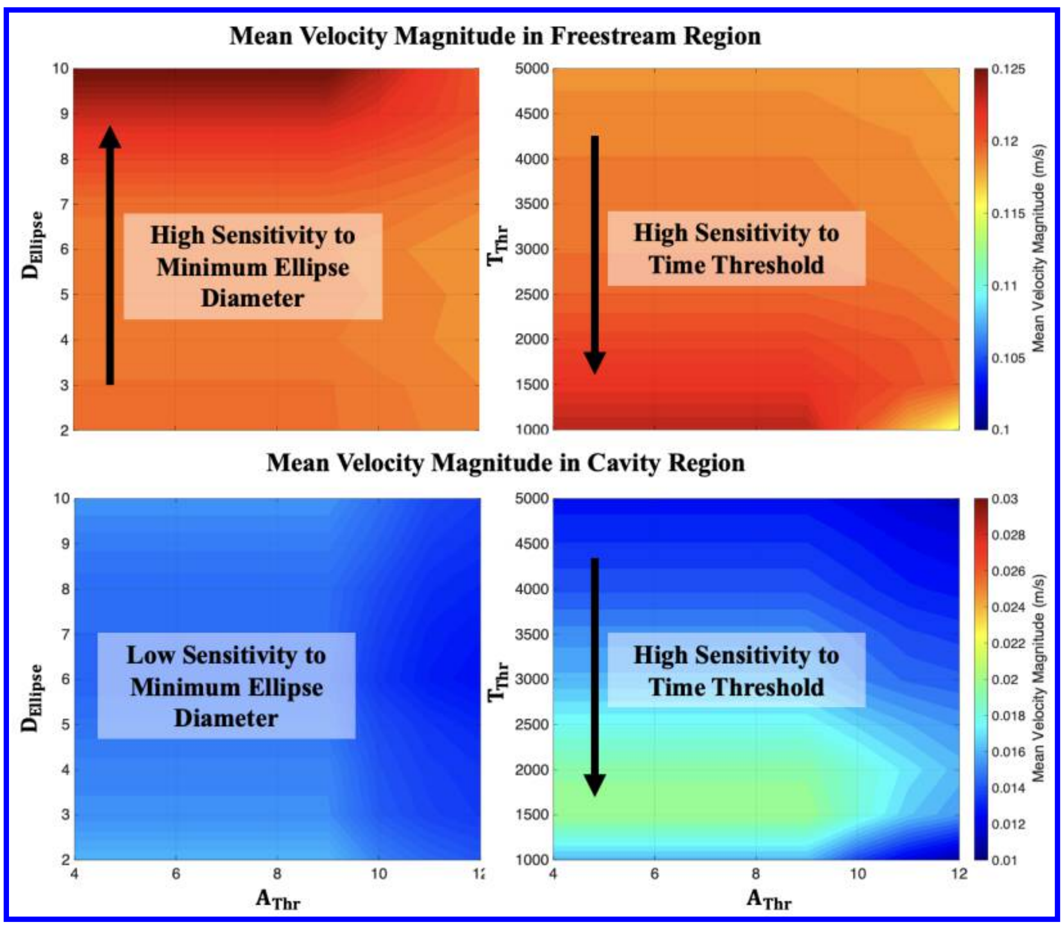

Figure 10: Accumulation Threshold vs Track Defining Thresholds in Both Freestream (top row) and Cavity (bottom row) Regions of Interest

When $D_{Ellipse}$ is small the difference between the base distance limit B_{Limit}, the minimum distance for an event to be 'close enough' to an existing track, and $D_{Ellipse}$ is positive and the search region can stretch along the particle path. Successive events from the same particle are likely to stay inside the ellipse, so tracks grow longer and remain smooth and uniform, which yields a stable mean velocity. When $D_{Ellipse}$ is large the difference becomes small or even negative so the search region does not grow and can even shrink along the motion direction. Small deviations in position or timing then push events outside the ellipse, the tracker repeatedly terminates and restarts tracks, and many short random trajectories appear. This changes the population of tracks that survive validation and produces the strong sensitivity of the freestream mean velocity to $D_{Ellipse}$. Changes in A_{Thr} threshold mostly remove the very shortest tracks and therefore have a much smaller influence on the mean.

The right panel in the top row shows that the freestream mean velocity is also sensitive to time threshold, T_{Thr}. T_{Thr} limits the maximum time gap allowed between consecutive events in a track. If this threshold is too small, valid freestream trajectories are broken into shorter segments or fail the temporal continuity test altogether, which alters the mix of tracks used to compute the mean velocity. As T_{Thr} increases, more complete trajectories are retained, and the mean velocity approaches a nearly constant value.

The bottom row presents the same parameter sweep for the cavity region. In the left panel the mean velocity depends only weakly on $D_{Ellipse}$. Inside the cavity the flow is slower, strongly sheared, and highly curved, and the particle images occupy a smaller portion of the field of view. Over the range of $D_{Ellipse}$ tested, the search region still captures similar recirculating paths, so the ensemble of accepted trajectories changes little, and the mean velocity remains nearly constant. In contrast, the cavity mean velocity is very sensitive to T_{Thr}, as seen in the bottom right panel. The slow and unsteady cavity motion produces larger and more irregular time gaps between events along a trajectory. When T_{Thr} is too small many of these slow cavity tracks fail the continuity requirement and are rejected. The remaining tracks are biased toward faster motions near the shear layer and the cavity opening, which increases the mean velocity. Larger T_{Thr} values allow the slower trajectories deeper in the cavity to survive, which lowers the mean. Together these trends show that $D_{Ellipse}$ mainly governs track quality in the nearly uniform freestream, while T_{Thr} is the dominant control parameter in the temporally irregular cavity flow.

Figure 11 explains in more detail why $D_{Ellipse}$ and A_{Thr} produce different mean values in the freestream contour. The central plot shows the $D_{Ellipse}$ versus A_{Thr} and the four surrounding three-dimensional plots show event tracks in a small freestream subregion for four extreme combinations of $D_{Ellipse}$ and A_{Thr}. In the upper left, $D_{Ellipse}$ is large and A_{Thr} is low. The minimum $D_{Ellipse}$ is therefore wide and the B_{Limit} - $D_{Ellipse}$ term leaves little room for elongation along the motion direction. The tracker starts many candidate tracks with similar initial size and frequently breaks them when new events fall just outside the search region. Because A_{Thr} is low, most of these short, irregular trajectories still pass the accumulation filter, which produces the large number of random looking tracks seen in this panel and contributes to the bias in the mean velocity at high $D_{Ellipse}$.

The lower left panel keeps A_{Thr} small but reduces $D_{Ellipse}$. The initial ellipse is now tighter and the difference in B_{Limit} - $D_{Ellipse}$ term is larger, so the search region can stretch along the particle path as the track grows. Successive events from the same particle remain inside the ellipse and the trajectories become long, smooth, and nearly parallel. The freestream is then represented by a dense set of uniform tracks, and the mean velocity is much closer to the expected value. The right column shows the effect of increasing A_{Thr}. In the lower right panel, $D_{Ellipse}$ is moderate and A_{Thr} is high. The track geometry is similar to the lower left case, but many of the shortest trajectories are removed because they do not meet the accumulation requirement. The remaining tracks are fewer in number but still uniform and well aligned, so the mean velocity changes only slightly. In the upper right panel, both $D_{Ellipse}$ and A_{Thr} are large. The wide initial ellipse again produces many irregular candidate tracks, but now most of them fail the combination of accumulation and time linearity checks and are discarded. Only a small number of long tracks survive, which makes the mean velocity more sensitive to the details of how those few trajectories are fitted and explains the stronger variation in the contour at high $D_{Ellipse}$ and high A_{Thr}.

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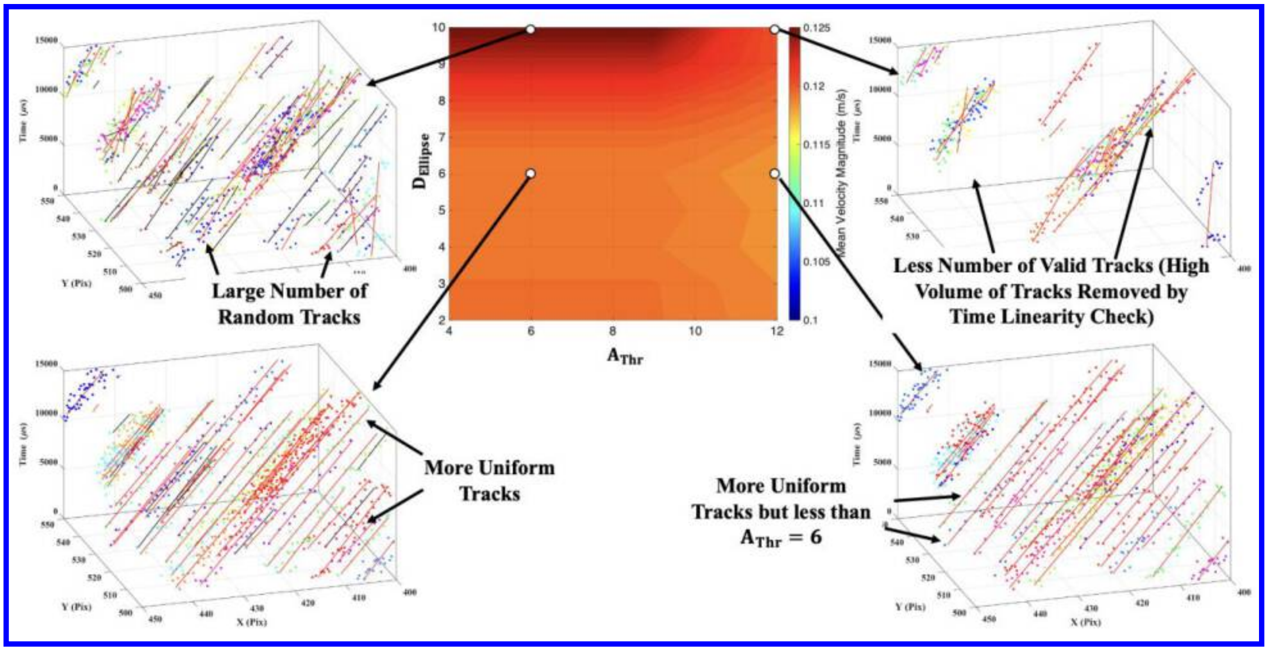

Figure 11 Effect of minimum ellipse diameter and accumulation threshold on freestream tracking. Center contour shows mean freestream velocity, surrounding 3D plots illustrate how varying $D_{Ellipse}$ and A_{Thr} changes the number and of valid tracks in the slow cavity flow.

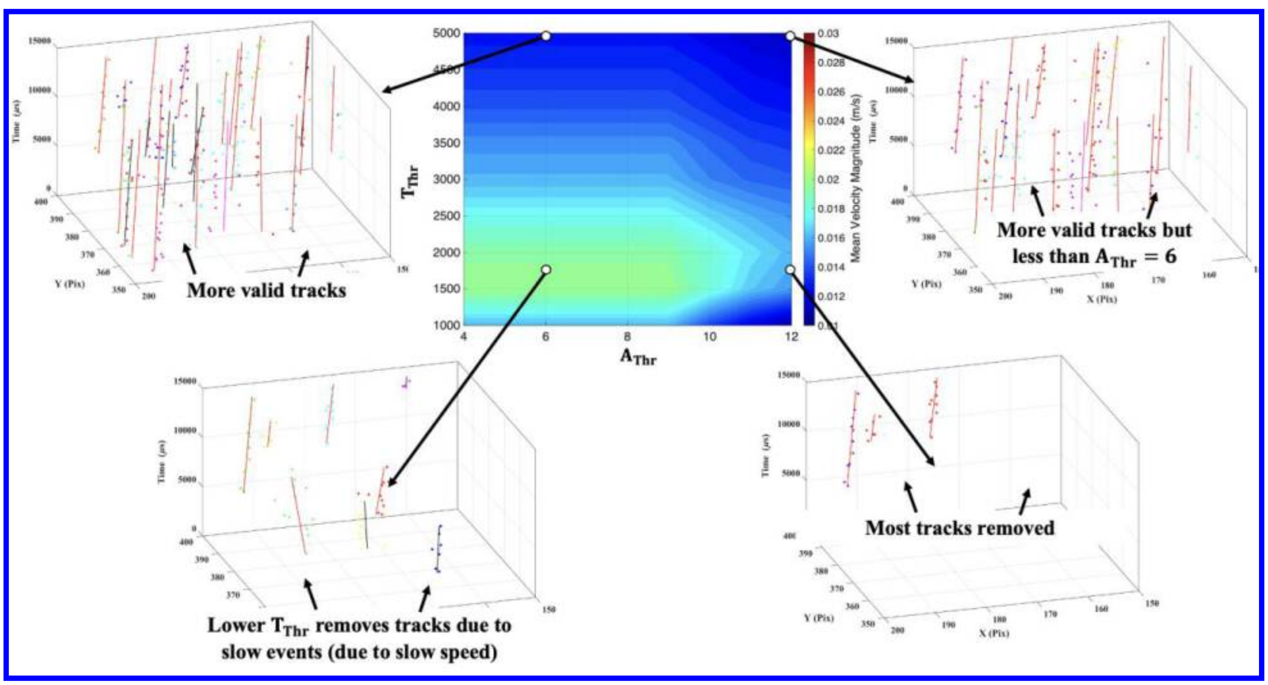

Figure 12 Effect of time threshold T_{Thr} and accumulation threshold A_{Thr} on cavity tracking. Center contour shows mean cavity velocity, surrounding 3D plots illustrate how varying T_{Thr} and A_{Thr} changes the number and of valid tracks in the slow cavity flow.

Figure 12 explains why the cavity mean velocity in Figure 10 is highly sensitive to the time threshold and only weakly dependent on the accumulation threshold. The central contour shows the mean cavity velocity as a function of T_{Thr} and A_{Thr}. In an ideal case every threshold combination would yield the same mean value, and the contour would be nearly uniform. Instead, decreasing T_{Thr} produces a clear drop in mean velocity because an increasingly large fraction of cavity tracks fails the temporal continuity requirement.

The surrounding three-dimensional plots show event tracks in a small portion of the cavity for four representative combinations of T_{Thr} and A_{Thr}. The upper left panel corresponds to a large T_{Thr} and a moderate A_{Thr}. The flow in the cavity is slow compared with the freestream, so events along a trajectory are separated by relatively long-time gaps. A generous T_{Thr} allows these gaps and most trajectories pass both the accumulation and time linearity checks, which results in many valid tracks. When T_{Thr} is reduced while keeping A_{Thr} fixed, as in the lower left panel, the maximum allowed gap between events becomes shorter than the typical inter event spacing for the slow cavity motion. Many trajectories are broken into fragments or fail the time linearity test, and a significant number of tracks is removed.

The upper right panel illustrates the effect of increasing A_{Thr} at a large T_{Thr}. Increasing A_{Thr} eliminates the shortest and weakest trajectories, so fewer tracks remain, but the surviving tracks are still long and representative of the cavity motion. The mean velocity changes only modestly. In contrast, when both T_{Thr} and A_{Thr} are large, as in the lower right panel, only a small subset of trajectories satisfies the combined requirements on minimum length and temporal continuity. Most of the slow cavity tracks are rejected and only a limited number of faster trajectories. This strong pruning explains the pronounced variation of the mean cavity velocity along the T_{Thr} axis in the central contour plot.

B. Velocity Sensitivity of Track Validation Thresholds

In Figure 13 the sensitivity of the freestream mean velocity to the time linearity threshold TL_{Thr} and the accumulation threshold A_{Thr} is examined. Once again, in an ideal case the mean velocity would not depend on either threshold and the central contour plot would show a single uniform color. Here the contour is nearly flat, which indicates that the mean freestream velocity is only weakly affected by the specific combination of TL_{Thr} and A_{Thr}. The surrounding three-dimensional plots help explain why the mean remains stable even though the track population changes.

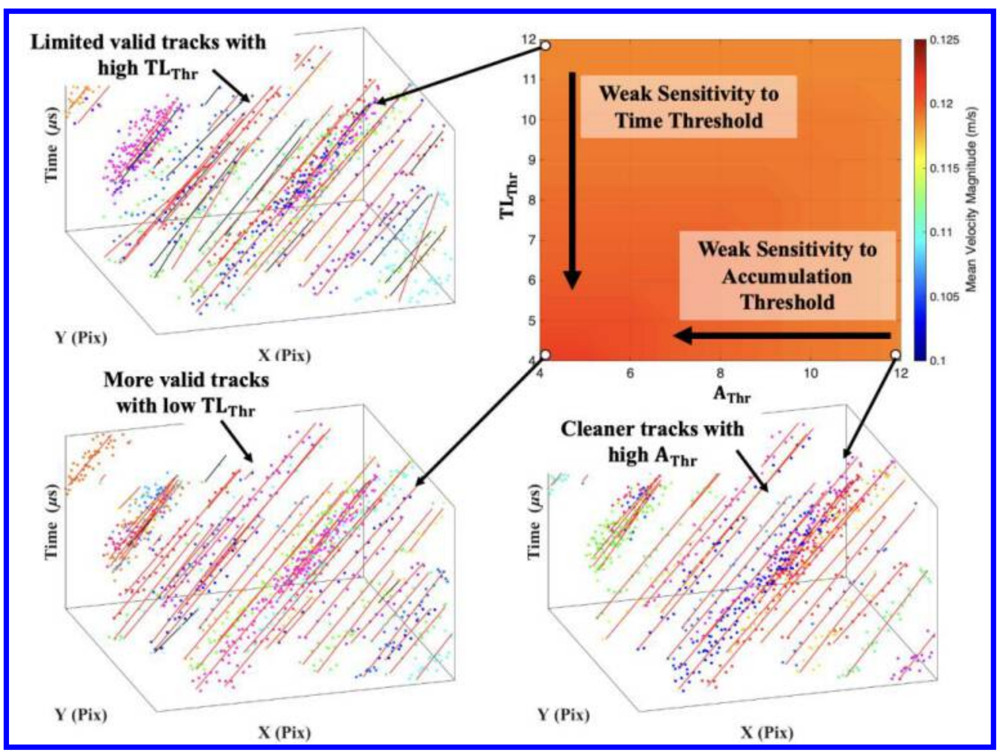

Figure 13 Sensitivity of freestream mean velocity to time linearity threshold TL_{Thr} and accumulation threshold A_{Thr}. Top right shows nearly uniform mean velocity, while surrounding 3D plots illustrate how varying TL_{Thr} and A_{Thr} mainly changes the number and cleanliness of valid tracks.

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The upper left panel corresponds to a large TL_{Thr} with a low A_{Thr}. The time linearity check is strict in this case and requires a long segment of nearly uniform inter event spacing, so only a limited number of trajectories pass the test. The surviving tracks are long and well aligned with the freestream direction, and although they are few in number they still represent the underlying uniform flow, so the mean velocity is accurate.

The lower left panel shows the effect of reducing TL_{Thr} while keeping A_{Thr} fixed. The temporal uniformity requirement is relaxed, and many more trajectories are accepted. Some of these additional tracks are noisier and contain small timing irregularities, but the dominant direction and speed are still set by the freestream motion. As a result, the mean velocity changes very little, even though the number of valid tracks increases.

The lower right panel illustrates the influence of increasing A_{Thr} at a moderate TL_{Thr}. Increasing A_{Thr} removes short tracks with only a few events and leaves a cleaner set of long trajectories. Again, these tracks all follow the same freestream direction, so the mean velocity remains essentially unchanged. Together these observations show that in freestream conditions the time linearity and accumulation thresholds primarily control the number and cleanliness of tracks, while having only a minor impact on the recovered mean velocity.

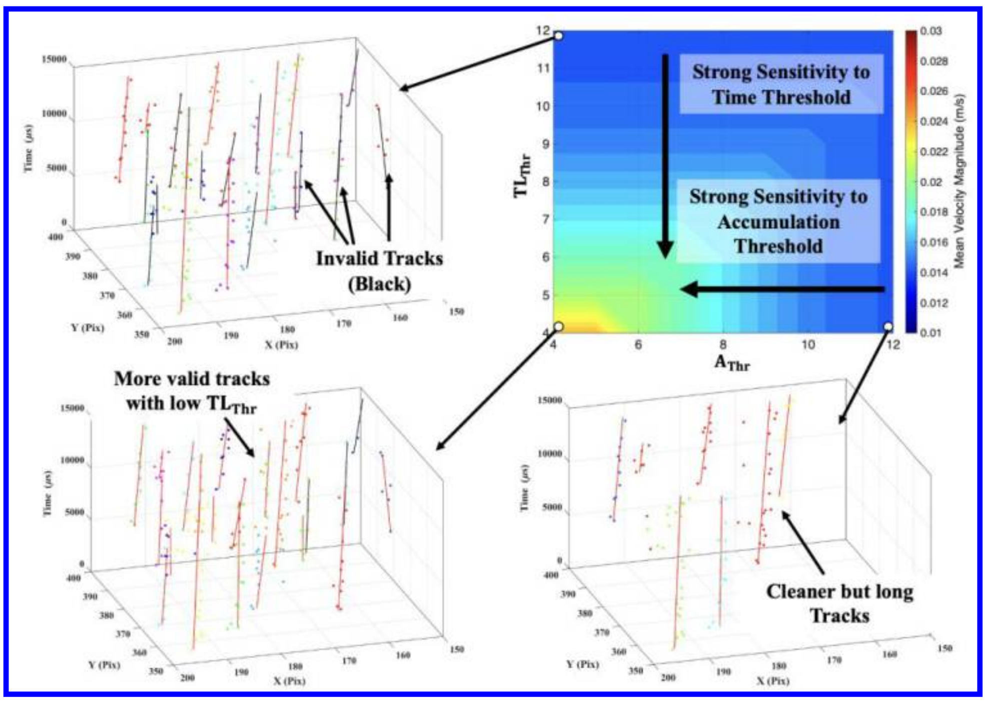

Figure 14 Sensitivity of cavity mean velocity to time linearity threshold $T_{L\mathrm{Thr}}$ and accumulation threshold A_{Thr}. Top right contour and surrounding 3D tracks highlight strong dependence on both validation thresholds.

In Figure 14 the sensitivity of the cavity mean velocity to the time linearity threshold TL_{Thr} and accumulation threshold A_{Thr} is examined. The cavity mean velocity magnitude shows a strong gradient in both directions, which indicates that the validation thresholds play a major role in deciding the accuracy of the velocity estimate.

The surrounding three-dimensional plots show event tracks in a small cavity region for three representative combinations of TL_{Thr} and A_{Thr}. In the upper left panel TL_{Thr} is large and the temporal linearity requirement is strict. Events in the cavity arrive with long and irregular time gaps because the flow is slow and unsteady. Many trajectories therefore fail the linearity test and are flagged as invalid, shown in black. Only a limited number of tracks survive, and these tend to correspond to the most regular motions, so the resulting mean velocity is strongly biased by this small subset. When TL_{Thr} is reduced while A_{Thr} is held fixed, as in the lower left panel, the algorithm accepts much shorter time linear segments. Many more trajectories pass validation, including tracks that sample both the slow core and the shear layer, and the mean velocity increases.

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The panel on the lower right illustrates the effect of increasing A_{Thr} at moderate TL_{Thr}. Raising A_{Thr}rejects tracks with only a few events, which often correspond to noisy or partially observed motions near the cavity edges. The remaining trajectories are longer and cleaner but are fewer in number, so the mean velocity is dominated by the slower, well resolved paths that remain. This tends to lower the mean compared with the low A_{Thr} case. Together, the contour and the track visualizations show that in the cavity the combination of TL_{Thr} and A_{Thr} does not simply change the number of tracks, it reshapes which parts of the slow, highly sheared flow are represented, which explains the strong sensitivity of the cavity mean velocity to these validation thresholds.

VII.Conclusion

This work presented a sensitivity analysis of event-based velocimetry using a weighted tracking algorithm applied to two configurations, a freestream and post stall wing flow and a freestream–cavity experiment. We examined three groups of parameters: camera bias settings, time segmentation, and the internal thresholds that govern track formation and validation. For the camera, the high pass filter bias strongly affected event density and thus computational cost. Higher HPF settings produced sparser event streams and reduced processing time by more than an order of magnitude, with only minor changes in median velocity once a sufficient event density was reached. In contrast, the time segment duration had a stronger influence on velocity estimates. Short segments slightly underpredicted velocity due to fragmented tracks, whereas long segments increased event overlap and produced a small positive bias in the nonlinear fits. The threshold study showed that the minimum ellipse diameter D_{Ellipse}, accumulation threshold A_{Thr}, and time thresholds T_{Thr} and TL_{Thr} control which trajectories survive. In the freestream, small D_{Ellipse} and moderate A_{Thr} and time thresholds yielded long, uniform tracks and stable mean velocities. In the slow cavity, the method was most sensitive to T_{Thr} and TL_{Thr}, which must be large enough to accommodate long inter event times without discarding genuine slow motions.

Overall, the results show that event-based velocimetry can provide reliable velocity estimates across both fast and slow regions of a flow, but only when sensor biases and tracking thresholds are tuned together. The sensitivity trends and track level diagnostics reported here offer practical guidelines for selecting those settings in future experiments. For freestream-type measurements, higher HPF biases, moderate time segments, small D_{Ellipse}, and moderate values of A_{Thr} and the time linearity threshold produce accurate velocities with reduced computational cost. For slow or strongly sheared regions such as the cavity interior, the tracker must be allowed longer temporal gaps, and the validation thresholds must be relaxed enough to retain genuinely slow trajectories while still filtering obvious noise.

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